

```

#include <windows.h>
#include <GL/glut.h>
#include <math.h>

const float PI = 3.14159265358979323846;

float ballX = 0.0f;
float ballY = 0.0f;
float ballRadius = 0.1f;
float xSpeed = 0.01f;
float ySpeed = 0.012f;

void drawCircle(float radius) {
    int numSegments = 100;
    glBegin(GL_POLYGON);
    for (int i = 0; i < numSegments; i++) {
        float angle = 2.0f * PI * i / numSegments;
        float x = radius * cos(angle);
        float y = radius * sin(angle);
        glVertex2f(x, y);
    }
    glEnd();
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);
    glLoadIdentity();

    glTranslatef(ballX, ballY, 0.0f);
    glColor3f(0.1f, 0.6f, 1.0f);
    drawCircle(ballRadius);

    glutSwapBuffers();
}

void update(int value) {

    ballX += xSpeed;
    ballY += ySpeed;

    if (ballX + ballRadius > 1.0f || ballX - ballRadius < -1.0f)
        xSpeed = -xSpeed;

    if (ballY + ballRadius > 1.0f || ballY - ballRadius < -1.0f)
        ySpeed = -ySpeed;

    glutPostRedisplay();
    glutTimerFunc(16, update, 0);
}

```

```
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(600, 600);
    glutInitWindowPosition(100, 100);
    glutCreateWindow("Bouncing Ball Animation");

    glClearColor(1.0, 1.0, 1.0, 1.0);
    glutDisplayFunc(display);
    glutTimerFunc(0, update, 0);

    glutMainLoop();
    return 0;
}
```